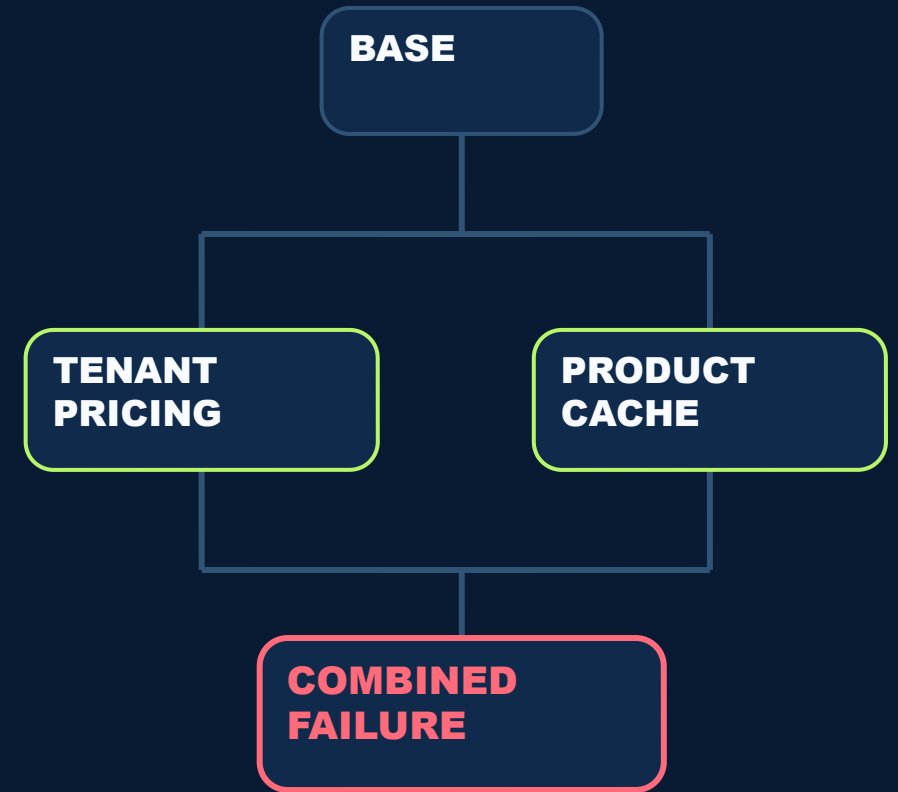


MergeWitness

A developer workflow for bugs that appear only when two green changes meet.



SYNTHETIC FIXTURE
Interaction witness verified

The failure hides behind green tests

Measured synthetic fixture. Original tests exit with code 0 in all four snapshots.

CHANGE A

Tenant-specific prices
Alpha: \$90 Beta: \$100

CHANGE B

Cache by product
Repeated requests avoid
source calls

1	Alpha requests SKU-1	\$90
2	Cache stores SKU-1	\$90
3	Beta requests SKU-1	\$90 ×
	Expected for Beta	\$100

Cache key omits the tenant

The ordinary suite can still pass because it checks each feature alone.

One invariant across four snapshots

Measured synthetic fixture. Three repetitions per snapshot; original tests exit with code 0 in all four.

Invariant: serving Alpha must not change Beta's price

Base

- Tests: exit 0
- Probe: 3× PASS

Change A

- Tests: exit 0
- Probe: 3× PASS

Change B

- Tests: exit 0
- Probe: 3× PASS

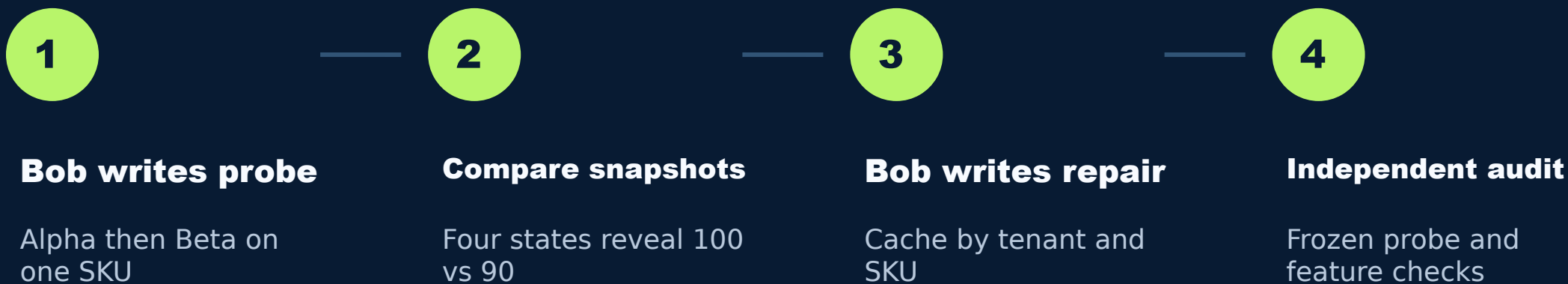
Combined

- Tests: exit 0
- **Probe: 3× FAIL**

Classification: interaction witness | Combined result: expected 100, observed 90

Bob probes and repairs the interaction

Three recorded IBM Bob tasks across two synthetic fixtures.



3 Bob sessions recorded. Second scenario: priority/cursor probe, combined fails 3/3.

A repair must keep both features

Measured synthetic fixture. Final audit passed at 17ed2d8c...d94e.

Before

cache[sku] = price

Beta receives Alpha's cached price

BOB

Bob-authored repair

Nested Map cache

Tenant and SKU remain distinct

Frozen interaction probe

● VERIFIED PASS

Tenant prices remain distinct

● VERIFIED PASS

Cache remains effective

● VERIFIED PASS

MERGEWITNESS

Evidence for the release decision

Measured tenant-cache fixture. A green suite missed a wrong customer price.

EXISTING SUITE

Tests pass.
Beta receives \$90
instead of \$100.

INTERACTION WITNESS

The same two
requests reproduce
the failure in the
combined change.

VERIFIED REPAIR

Tenant prices stay
distinct.
Repeated requests
still use the cache.

Hold the failing merge. Review the repair with both features preserved.

These synthetic checks cover the executed scenarios. A passing probe cannot certify every merge as safe.

Live demo: <https://lawliet8886.github.io/MergeWitness/>

Code and evidence: <https://github.com/lawliet8886/MergeWitness>